Duplicate Leakage in Stacking-Based Heart Attack Prediction: Reproduction, Critical Analysis and a Leakage-Aware Convex Stacking Method

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Deliverable	Link
Video presentation (≤ 5 min)	[YouTube (unlisted) link]
Code, data and README	https://github.com/Akshit307/mlhd

Abstract

Bhagat, Sharma and Agarwal [1] report that stacking six classifiers (LR, DT, RF, XGBoost, NB, KNN) raises heart-attack prediction accuracy from 92.68% to **98.53%** on a public 1,025-record heart-disease dataset. We reproduce their pipeline and obtain 99.43% mean accuracy over 30 random 80/20 splits, confirming the headline result. A data audit, however, shows that only **302 of the 1025 records are unique**: 723 are exact duplicates, and on average 97.3% of test records have an identical copy in the training set. When the identical pipeline is evaluated on unique patients, stacking accuracy falls to 84.6% and the decision tree falls from 99.2% to 76.0%; the reported gains are largely an artefact of leakage. We then propose **LACS** (Leakage-Aware, Clinically-encoded, Convex Stacking), which combines (i) a patient-level evaluation protocol, (ii) clinically correct handling of undocumented missing-value codes and nominal features, and (iii) a convex, shrinkage-regularised meta-learner over a diverse, regularised base pool. Under 10×5 repeated stratified cross-validation with identical folds, LACS improves on the paper's stacking in all seven metrics (accuracy 83.1% → 84.6%, F1 0.850 → 0.863, AUC 0.905 → 0.913, log-loss 0.399 → 0.378). The gains are consistent across folds (Wilcoxon $p = 0.005$ for accuracy) but modest relative to between-fold variance (corrected resampled t-test $p = 0.37$). Ablations show that a clinically-encoded, regularised logistic regression alone is as good as any ensemble on 302 patients. The realistic ceiling for this data is therefore about 85% accuracy and 0.91 AUC, not 98.5%.

1. Introduction and Research Problem

Cardiovascular disease is the leading cause of death worldwide, and early, cheap risk screening from routinely collected clinical variables (age, sex, chest-pain type, resting blood pressure, cholesterol, ECG findings, exercise test results) remains an active machine-learning problem [1], [11], [12], [13]. A large body of work reports increasingly high accuracies using hybrid and ensemble methods on UCI-derived heart-disease data. Stacked generalisation [4], in which a meta-learner combines the predictions of several base classifiers, is a popular route to such gains [1], [12], [13].

Bhagat *et al.* [1] (the selected paper) study the 1,025-record Kaggle release [3] of the UCI Heart Disease data [2]. They pre-process the data (missing-value filling, standardisation, categorical encoding), split it into training and test sets, train six classifiers, and stack them with a meta-classifier. The **research gap** they address is that single classifiers plateau at around 90 to 93% accuracy; their **claim** is that stacking lifts accuracy to 98.53%, outperforming every individual model. This report (Part 1) reproduces that claim, critically tests whether it reflects generalisation to new patients, and (Part 2) proposes and rigorously evaluates a method that addresses the limitations found.

2. Summary of the Selected Paper and State of the Art

Dataset. 1,025 records × 13 clinical features + binary target (heart disease present / absent), 526 positive and 499 negative [1], [3].

Methods. Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGB) [9], Gaussian Naive Bayes (NB) and K-Nearest Neighbours (KNN) as base learners; a stacking ensemble in which a meta-classifier is trained on the base learners' predictions [4]. Evaluation uses accuracy, precision, recall and F1 on a single hold-out test set.

Reported results. RF and DT 92.68%, XGB 90.73%, stacking 98.53% accuracy [1]. All three values are exact multiples of 1/205 (190/205, 186/205, 202/205), which implies a 205-record test set, i.e. an 80/20 split of 1,025 records.

State of the art. Related work on UCI heart-disease data spans feature-selection pipelines combined with classical classifiers [11], hybrid random forest with a linear model (HRFLM) [12], optimised stacked SVMs [13] and majority-voting or bagging/boosting ensembles [14]. Most of these works report results from a single split or a single k-fold run, often without confidence intervals or significance tests, which makes cross-paper comparison fragile [7], [15]. Kapoor and Narayanan [6] document data leakage as a pervasive cause of over-optimistic results in ML-based science, including medicine. Leakage of this kind is exactly what our reproduction uncovers.

3. Part 1: Reproduction of the Published Results

3.1 Implementation and assumptions

The full paper text does not report hyperparameters, the meta-learner, random seeds or which variables were treated as categorical. We implemented the pipeline in scikit-learn [8] and XGBoost [9] and made the following documented assumptions (Table I). The dataset has no NaN values, so the 'missing-value filling' step has no effect on the file as distributed.

TABLE I. Reproduction protocol and assumptions.

Aspect	Paper states	Our implementation (assumption)
Data	1,025 patients, 14 attributes	Kaggle heart.csv [3]; identical shape and class balance (526/499)
Split	train/test hold-out	Stratified 80/20 (205 test records, inferred from reported accuracies); seed 42, plus 30 seeds for variance
Pre-processing	fill missing, standardise, encode	No NaNs present; StandardScaler for scale-sensitive models (LR, KNN); codes used as given
Base models	LR, DT, RF, XGB, NB, KNN	Library defaults (hyperparameters not reported)
Stacking	meta-classifier on base predictions	sklearn StackingClassifier, LR meta-learner on 5-fold out-of-fold probabilities
Metrics	Accuracy, Precision, Recall, F1	Same + ROC-AUC, log-loss and Brier score

3.2 Reproduced results

TABLE II. Paper protocol (1,025 records, 80/20 split). Precision, recall, F1 and AUC are 30-split means. n/r = not reported in the paper's abstract.

Model	Paper Acc.	Acc. (seed 42)	Acc. 30 splits (mean ± SD)	Precision	Recall	F1	AUC
LR	n/r	80.98%	83.72% ± 2.44	0.807	0.899	0.850	0.913
DT	92.68%	98.54%	99.17% ± 0.99	0.996	0.988	0.992	0.992
RF	92.68%	100.00%	99.58% ± 0.85	0.996	0.996	0.996	1.000
XGB	90.73%	100.00%	99.48% ± 0.96	0.995	0.995	0.995	0.997
NB	n/r	82.93%	81.61% ± 2.00	0.799	0.858	0.827	0.899
KNN	n/r	86.34%	84.72% ± 2.09	0.848	0.858	0.852	0.956
Stacking	98.53%	100.00%	99.43% ± 0.94	0.996	0.993	0.994	0.999

The headline claim reproduces: stacking reaches 99.43% mean accuracy (paper 98.53%) and is at or near the top of all models. Two discrepancies stand out. (1) Our tree-based models (DT, RF, XGB) score ≈99 to 100%, far above the paper's 90.7 to 92.7%. This is consistent with the paper having used constrained trees (e.g. limited depth) or a different split, which is plausible because hyperparameters were not reported. (2) In our runs stacking does not beat RF and XGB by a clear margin; all three sit within about one standard deviation of 100%. The variation across 30 splits (SD ≈ 1 percentage point) shows that a single split cannot resolve differences of the size the paper reports.

3.3 Critical analysis: why is the accuracy so high?

Duplicate-record leakage. An audit of the data (notebook §0) shows **723 exact duplicate rows; only 302 records are unique** (close to the 303-patient Cleveland cohort [2], [10]). The 1,025-record file is effectively the ~300 Cleveland patients repeated about 3.4 times. Under the paper's random 80/20 split, 97.3% of test records (min 92.2%, over 30 splits) have an identical feature vector in the training set. Any model that can memorise, such as an unpruned tree, RF, XGB or a stacker built on them, simply looks the answer up. This is textbook train/test contamination [5], [6]. It explains why the most memorising models look best and why stacking appears to 'fix' the errors.

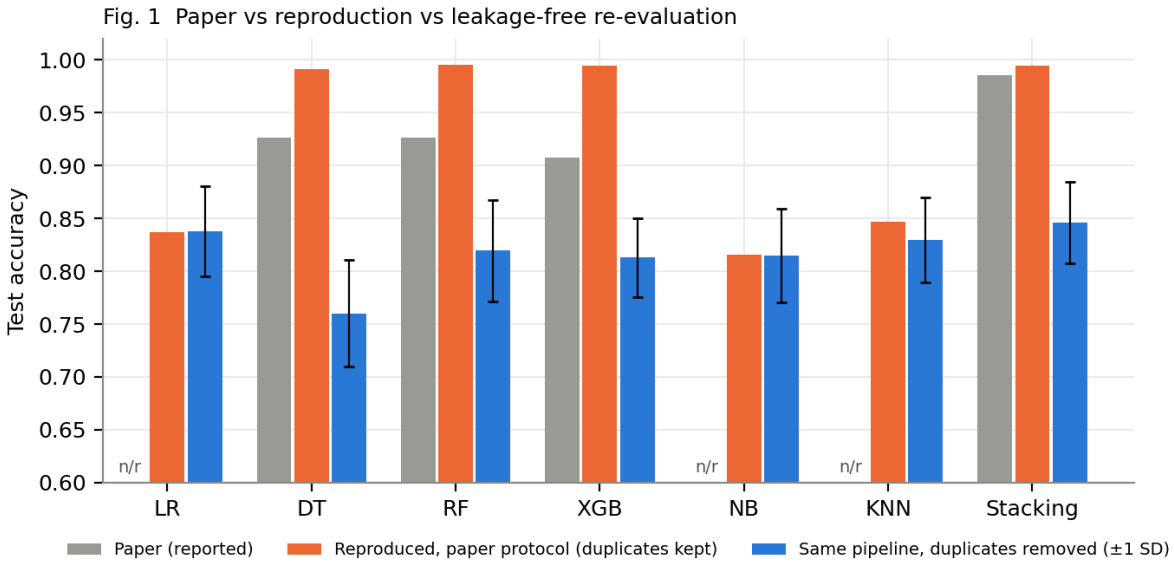


Fig. 1. Accuracy reported in the paper (grey), reproduced with the paper's protocol (orange), and the *identical* pipeline re-run on the 302 unique records (blue, 30 random 80/20 splits, ±1 SD).

Re-running the *unchanged* pipeline on de-duplicated data (Fig. 1, Fig. 2) removes the illusion. Stacking falls from 99.4% to 84.6%, RF from 99.6% to 82.0%, XGB to 81.3% and DT to 76.0%. The models that cannot memorise exact points are unaffected: LR stays at 83.8% and NB at 81.5%. This differential collapse is the signature of leakage rather than genuine skill. The ranking also flips: on unique patients the simple LR is statistically indistinguishable from stacking, and DT, which the paper ranks joint-first, is the worst model.

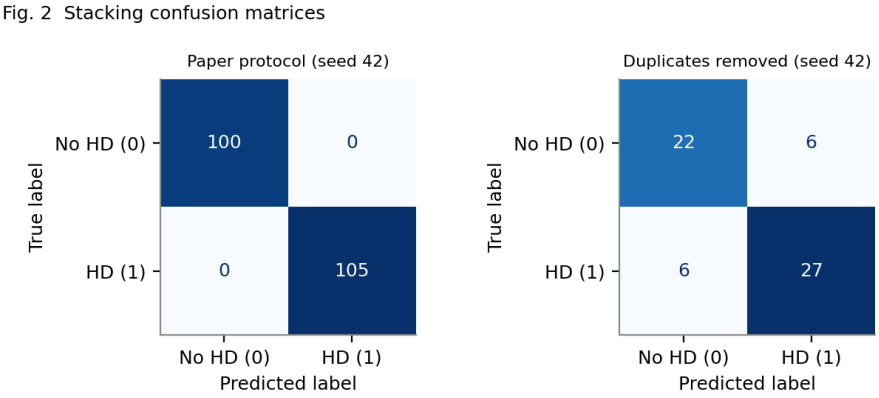


Fig. 2. Stacking confusion matrices, seed 42. Left: paper protocol (205/205 correct). Right: identical pipeline on unique patients (61 test patients).

Further methodological issues. (a) *Undocumented codes.* The file contains ca = 4 and thal = 0, values that do not exist in the UCI codebook [2]; they correspond to the original data's missing values and were silently recoded, so 'filling missing values' was never actually applied. (b) *Nominal codes treated as numbers.* Chest-pain type, resting ECG, ST slope and thalassaemia are unordered categories, yet as integer codes a linear or distance-based model assumes, for example, that cp = 3 is 'three times' cp = 1. (c) *Single hold-out, no uncertainty.* Results are point estimates from one split with no confidence intervals or tests [7], [15]. (d) *Unconstrained meta-learner.* An LR meta-learner on six highly correlated probability outputs can assign large, unstable, even negative weights. Breiman [16] showed that constraining stacking weights to be non-negative is important for stable stacked predictors. (e) *Probability quality is not assessed*, although calibrated risk is what matters clinically.

4. Part 2: Proposed Method: LACS

4.1 Motivation

Part 1 shows the paper's key limitation is not the choice of classifier but **how the data are prepared and evaluated**. A fair improvement must therefore (1) be measured on unseen *patients*, (2) use the clinical variables correctly, and (3) combine models in a way that is robust when only ~240 patients are available for training. LACS addresses each of these in turn.

4.2 Method

(1) Leakage-aware protocol. Records are de-duplicated to patient level before any split. Duplicates add no information and would act as arbitrary sample weights. All fitting (imputation, scaling, encoding, hyperparameter tuning and meta-weight estimation) happens inside the training fold only. Evaluation uses 10×5 repeated stratified cross-validation (50 folds), and every model sees the *same* folds, so comparisons are paired.

(2) Clinically-typed encoding. ca = 4 and thal = 0 are mapped back to missing and imputed (median / mode) from the training fold. The nominal variables cp, restecg, slope and thal are one-hot encoded. Continuous variables (age, trestbps, chol, thalach, oldpeak, ca) are standardised, and binary flags (sex, fbs, exang) are passed through.

(3) Diverse, regularised base pool. L2-logistic regression with its regularisation strength tuned by inner 5-fold CV on log-loss (12 values of C in $[10^{-3}, 10^{2}]$); RBF-SVM (C = 0.5, Platt-scaled); distance-weighted KNN (k = 15); Gaussian NB; Random Forest and Extra-Trees [17] (500 trees, min leaf = 3); and shallow XGBoost (depth 2, 300 trees, $\eta = 0.05$, subsampling 0.8, $\lambda = 2$). Settings favour low variance, as suits $n \approx 240$, and were fixed *a priori* rather than tuned on the evaluation folds.

(4) Convex, shrunk meta-learner. Out-of-fold probabilities $p_k(x)$ of the K base models are obtained by inner 5-fold CV on the training fold. The meta-learner is a convex combination $p(x) = \Sigma_k\, w_k\, p_k(x)$, with $w_k \geq 0$ and $\Sigma\, w_k = 1$. The weights minimise out-of-fold log-loss (softmax parameterisation, L-BFGS) and are then shrunk toward the uniform average: $w = (1 - \lambda)\, w^* + \lambda/K$ with $\lambda = 0.5$. The non-negativity constraint follows Breiman [16] and the probability-level combination follows Ting and Witten [18]. Shrinkage guards against weight-estimation noise with small n, and the output stays a calibrated-looking probability rather than a logit of correlated inputs. Base models are finally refit on the whole training fold.

TABLE III. How LACS differs from the original method.

Component	Paper [1]	LACS (proposed)
Evaluation unit	records (duplicates cross the split)	unique patients; 10×5 repeated stratified CV; paired folds
Missing codes	ca = 4, thal = 0 used as real values	mapped to NaN, imputed inside training fold
Nominal features	integer codes	one-hot encoded
Base learners	6 default models incl. unpruned DT	7 regularised, diverse models; LR C tuned by inner CV
Meta-learner	unconstrained LR on base outputs	convex non-negative weights (sum 1), shrunk to uniform
Reporting	single split, point estimates	mean ± SD over 50 folds, corrected t-test, Wilcoxon, calibration

4.3 Experimental protocol

Data: 302 unique patients (164 positive, 138 negative). Folds: RepeatedStratifiedKFold(5 splits × 10 repeats, seed 42), giving ≈ 241 training and 61 test patients per fold. Compared models, all evaluated on identical folds: the **paper stacking** baseline (exact Part 1 pipeline); the **paper's best single model** (RF); and four **ablations** that isolate each LACS component: paper stacking with clinical encoding (encoding only), uniform soft-voting of the LACS pool ($\lambda = 1$), convex weights without shrinkage ($\lambda = 0$), and the tuned clinical LR alone. Metrics: accuracy, precision, recall, F1, ROC-AUC (threshold 0.5 for label metrics), log-loss and Brier score. Significance: the Nadeau and Bengio corrected resampled t-test [7], which accounts for overlapping training sets in repeated CV, together with the Wilcoxon signed-rank test [15]. The Wilcoxon test is reported for completeness but is known to be anti-conservative on overlapping CV folds, so the corrected t-test is the primary criterion.

4.4 Results

TABLE IV. Leakage-free 10×5 repeated CV on 302 unique patients (mean ± SD over 50 identical folds).

Model	Accuracy	Precision	Recall	F1	AUC	Log-loss ↓	Brier ↓
Paper stacking (baseline)	0.831 ± 0.039	0.826 ± 0.048	0.879 ± 0.050	0.850 ± 0.033	0.905 ± 0.031	0.399 ± 0.060	0.124 ± 0.023
Paper best single model (RF)	0.818 ± 0.045	0.815 ± 0.049	0.865 ± 0.061	0.837 ± 0.040	0.903 ± 0.031	0.407 ± 0.051	0.128 ± 0.018
A1: paper stacking + clinical encoding	0.836 ± 0.038	0.832 ± 0.041	0.878 ± 0.057	0.853 ± 0.034	0.907 ± 0.029	0.389 ± 0.051	0.120 ± 0.020
A2: LACS pool, uniform soft vote (λ = 1)	0.844 ± 0.031	0.832 ± 0.036	0.898 ± 0.057	0.862 ± 0.029	0.912 ± 0.028	0.379 ± 0.048	0.118 ± 0.017
A3: LACS pool, convex, no shrinkage (λ = 0)	0.849 ± 0.032	0.839 ± 0.035	0.896 ± 0.056	0.866 ± 0.030	0.912 ± 0.029	0.379 ± 0.050	0.118 ± 0.018
A4: clinical LR only	0.855 ± 0.030	0.843 ± 0.037	0.904 ± 0.052	0.871 ± 0.027	0.916 ± 0.029	0.372 ± 0.049	0.115 ± 0.018
LACS (proposed, λ = 0.5)	**0.846 ± 0.030**	**0.835 ± 0.034**	**0.898 ± 0.057**	**0.863 ± 0.029**	**0.913 ± 0.029**	**0.378 ± 0.049**	**0.117 ± 0.018**

TABLE V. Paired differences (LACS minus other) over 50 folds, with corrected resampled t-test [7] and Wilcoxon p-values. Positive Δ is better, except for log-loss.

LACS vs.	Δ Acc.	p (corr. t)	p (Wilc.)	Δ F1	p (corr. t)	Δ AUC	p (corr. t)	Δ Log-loss	p (corr. t)
Paper stacking (baseline)	+0.0149	0.369	0.005	+0.0133	0.373	+0.0074	0.436	-0.0207	0.251
Paper best single model (RF)	+0.0288	0.150	0.000	+0.0262	0.151	+0.0101	0.253	-0.0296	0.102
A1: paper stacking + clinical encoding	+0.0106	0.459	0.014	+0.0106	0.445	+0.0060	0.379	-0.0114	0.242
A2: LACS pool, uniform soft vote (λ = 1)	+0.0020	0.790	0.719	+0.0014	0.839	+0.0004	0.875	-0.0009	0.803
A3: LACS pool, convex, no shrinkage (λ = 0)	-0.0030	0.774	0.257	-0.0022	0.814	+0.0011	0.627	-0.0014	0.701
A4: clinical LR only	-0.0086	0.458	0.028	-0.0077	0.460	-0.0033	0.452	+0.0057	0.408

Fig. 3 Leakage-free 10×5 repeated CV (302 unique patients, identical folds)

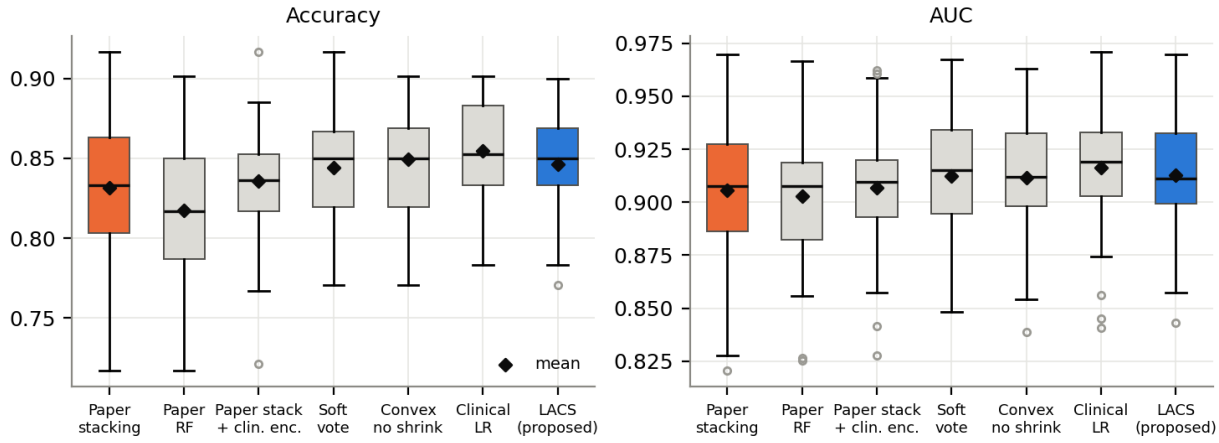


Fig. 3. Per-fold accuracy and AUC over 50 identical CV folds (black diamond = mean). Orange: paper stacking; blue: LACS; grey: comparators and ablations.

Fig. 4 Discrimination, calibration and learned ensemble weights

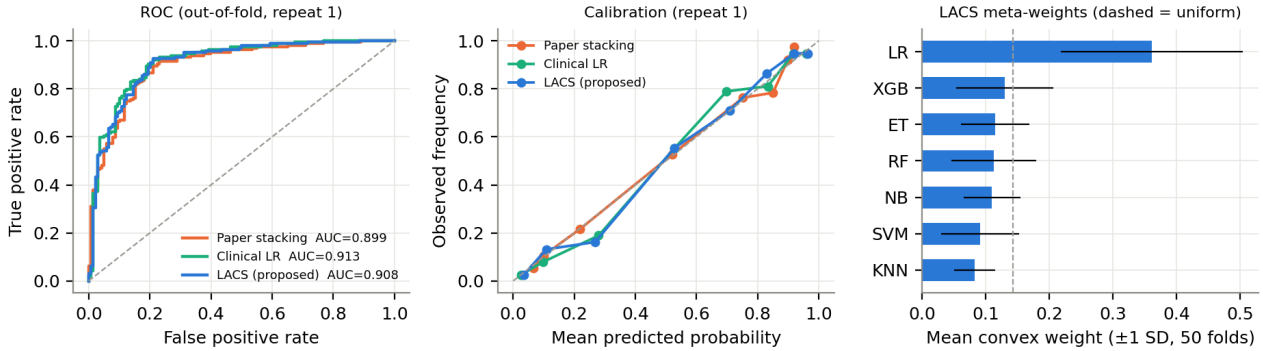


Fig. 4. Left: pooled out-of-fold ROC curves for one full CV repeat. Centre: reliability diagram. Right: mean LACS meta-weights across the 50 outer folds (dashed line = uniform 1/7).

4.5 Analysis and discussion

LACS vs. the reproduced baseline. On unseen patients, LACS improves on the paper's stacking in every metric: accuracy 83.14% → 84.63%, recall 0.879 → 0.898, F1 0.850 → 0.863, AUC 0.905 → 0.913, log-loss 0.399 → 0.378 and Brier 0.124 → 0.117. It also has a smaller fold-to-fold spread (accuracy SD 0.030 vs 0.039). LACS is more accurate in 29 of 50 folds and less accurate in 10, and has higher AUC in 30 folds. The improvement is consistent (Wilcoxon $p \leq 0.005$ for accuracy, F1 and log-loss). However, it is not significant under the conservative corrected t-test ($p = 0.37$), because the gain (≈ 1.5 points) is small relative to the ≈ 4-point between-fold variance that 61-patient test folds impose. The higher recall matters clinically: missed disease is the costlier error. Against the paper's best single model (RF), the gain is larger (+2.9 points accuracy; Brier corrected $p = 0.024$).

What each component contributes (ablations). Clinical encoding alone (A1) recovers just under a third of the gain (83.57%), and it improves log-loss, showing that correct treatment of nominal and missing codes helps even the original ensemble. Replacing the paper's pool and meta-learner with the regularised pool and a convex combiner (A2, A3, LACS) adds the rest. The three combiner variants are statistically indistinguishable (all corrected $p > 0.6$). The shrinkage in LACS gives the lowest log-loss and variance of the three but not the highest accuracy (A3: 84.93%). The learned weights (Fig. 4, right) put ≈ 0.36 of the mass on LR and spread the rest almost evenly. This mirrors the finding below: the linear signal dominates.

An honest negative result. The tuned, clinically-encoded LR alone (A4) scores 85.49% accuracy and 0.916 AUC. That is numerically slightly *better* than LACS, though not significantly so under the corrected test ($p = 0.46$). With 13 largely additive risk factors and ~240 training patients, ensembling cannot find much non-linear structure beyond a well-specified linear model. The practical lesson is that for this problem the large apparent benefit of stacking in [1] comes from the evaluation, not from the model. The main value of LACS over the original method is correct and robust handling of the data, with the ensemble acting as insurance: its weights automatically fall back toward the strongest learner.

Comparison with the paper's reported numbers. The paper's 98.53% cannot be compared directly with any leakage-free number. As a sanity check, running LACS under the paper's leaky protocol (10 seeds) gives 99.85% accuracy vs 99.56% for the paper's stacking. Both are at the ceiling, so the leaky protocol cannot discriminate between methods at all. On a valid protocol, the realistic performance for this cohort is ≈ 83 to 86% accuracy and ≈ 0.91 AUC.

Limitations and future work. (i) 302 patients from a single site: external validation on the Hungarian, Swiss and Long Beach VA cohorts [2] is the natural next step. (ii) The base-learner hyperparameters were fixed a priori to limit tuning cost; full nested tuning may add a little. (iii) The 0.5 decision threshold was not optimised for the asymmetric clinical cost of false negatives. A cost-sensitive threshold chosen inside the training fold would raise recall further. (iv) The Kaggle release binarises the UCI 0 to 4 diagnosis without documenting its re-coding, which limits clinical interpretation of individual coefficients.

5. Conclusion

We reproduced the stacking results of Bhagat *et al.* [1] (99.43% vs 98.53%) and showed that they rest on a dataset in which 723 of 1025 records are duplicates, so that ≈ 97% of test records are seen during training. Evaluated on unique patients, the same pipeline achieves 84.6%. The proposed LACS method, with leakage-aware evaluation, clinically correct encoding and a convex, shrunk meta-learner over regularised base models, consistently improves on the baseline in all metrics under a rigorous, paired 10×5 CV protocol (84.6% accuracy, 0.913 AUC). Ablations show that most of the achievable gain comes from treating the data correctly, and that a well-specified logistic regression is a hard-to-beat baseline for this problem. The broader lesson is that on small medical tabular datasets, duplicate audits and patient-level splits should come before model complexity.

Reproducibility Statement

All code, the dataset and a README are provided (link on page 1). `SIT307_HD_Heart_Stacking.ipynb` regenerates every table and figure from `data/heart.csv` with fixed seeds (≈ 25 min on a laptop CPU); `heartml.py` contains the models, LACS and the statistical tests; `make_report.py` rebuilds this PDF from the saved results. Environment: Python 3.11, scikit-learn 1.8.0, XGBoost 3.2.0, pandas, SciPy, Matplotlib (pinned in requirements.txt).

Acknowledgement of Generative AI Use

In accordance with Deakin University's academic integrity requirements, I acknowledge the use of Claude (Anthropic) as an assistant in this task. It was used to help select the paper, write and debug the Python implementation, run the experiments, and draft and format this report. I reviewed, ran and verified all code and results, and the analysis, conclusions and final wording are my own responsibility. *[Edit this statement so that it accurately reflects how you used GenAI.]*

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Appendix A: Additional results

TABLE A1. Paper protocol, 30 random stratified 80/20 splits of the 1,025 records (leaky).

Model	Acc. mean ± SD	Prec. mean ± SD	Recall mean ± SD	F1 mean ± SD	AUC mean ± SD
LR	0.8372 ± 0.0244	0.8066 ± 0.0298	0.8994 ± 0.0351	0.8498 ± 0.0225	0.9135 ± 0.0163
DT	0.9917 ± 0.0099	0.9956 ± 0.0114	0.9883 ± 0.0147	0.9919 ± 0.0097	0.9918 ± 0.0099
RF	0.9958 ± 0.0085	0.9957 ± 0.0114	0.9962 ± 0.0099	0.9959 ± 0.0082	0.9995 ± 0.0011
XGB	0.9948 ± 0.0096	0.9948 ± 0.0144	0.9952 ± 0.0132	0.9949 ± 0.0093	0.9969 ± 0.0076
NB	0.8161 ± 0.0200	0.7987 ± 0.0225	0.8581 ± 0.0339	0.8268 ± 0.0196	0.8989 ± 0.0180
KNN	0.8472 ± 0.0209	0.8485 ± 0.0384	0.8578 ± 0.0368	0.8519 ± 0.0188	0.9561 ± 0.0116
Stacking	0.9943 ± 0.0094	0.9956 ± 0.0114	0.9933 ± 0.0123	0.9944 ± 0.0092	0.9989 ± 0.0025

TABLE A2. Identical paper pipeline, 30 random stratified 80/20 splits of the 302 unique records.

Model	Acc. mean ± SD	Prec. mean ± SD	Recall mean ± SD	F1 mean ± SD	AUC mean ± SD
LR	0.8377 ± 0.0425	0.8313 ± 0.0512	0.8838 ± 0.0591	0.8549 ± 0.0372	0.9069 ± 0.0345
DT	0.7601 ± 0.0505	0.7859 ± 0.0538	0.7697 ± 0.0696	0.7757 ± 0.0494	0.7593 ± 0.0510
RF	0.8197 ± 0.0479	0.8203 ± 0.0484	0.8566 ± 0.0646	0.8367 ± 0.0448	0.9051 ± 0.0354
XGB	0.8131 ± 0.0372	0.8217 ± 0.0474	0.8414 ± 0.0620	0.8293 ± 0.0345	0.8900 ± 0.0340
NB	0.8148 ± 0.0444	0.8212 ± 0.0413	0.8434 ± 0.0764	0.8301 ± 0.0448	0.8973 ± 0.0376
KNN	0.8295 ± 0.0403	0.8281 ± 0.0490	0.8697 ± 0.0591	0.8464 ± 0.0350	0.8852 ± 0.0339
Stacking	0.8459 ± 0.0384	0.8404 ± 0.0446	0.8869 ± 0.0595	0.8613 ± 0.0353	0.9102 ± 0.0330

Data audit. rows = 1025; exact duplicate rows = 723; unique rows = 302; unique feature vectors with conflicting labels = 0. Mean share of test rows with an exact copy in the training set under the paper protocol: 97.3% (range 92.2% to 100.0% over 30 splits).